

Overview of Principal component analysis

Subject: Principal component analysis

1.

Further considerations The singular values (in Σ) are the square roots of the eigenvalues of the matrix $X^T X$. Each eigenvalue is proportional to the portion of the "variance" (more correctly of the sum of the squared distances of the points from their multidimensional mean) that is associated with each eigenvector. The sum of all the eigenvalues is equal to the sum of the squared distances of the points from their multidimensional mean. PCA essentially rotates the set of points around their mean in order to align with the principal components. This moves as much of the variance as possible (using an orthogonal transformation) into the first few dimensions. The values in the remaining dimensions, therefore, tend to be small and may be dropped with minimal loss of information (see below). PCA is often used in this manner for dimensionality reduction. PCA has the distinction of being the optimal orthogonal transformation for keeping the subspace that has largest "variance" (as defined above). This advantage, however, comes at the price of greater computational requirements if compared, for example, and when applicable, to the discrete cosine transform, and in particular to the DCT-II which is simply known as the "DCT". Nonlinear dimensionality reduction techniques tend to be more computationally demanding than PCA. PCA is sensitive to the scaling of the variables. Mathematically this sensitivity comes from the way a rescaling changes the sample covariance matrix that PCA diagonalises. Let $\mathbf{X}_c$ be the *centered* data matrix (n rows, p columns) and define the covariance

2.

$$Q(\mathrm{PC}(j), \mathrm{PC}(k)) \propto (\mathbf{X} \mathbf{w}(j))^T (\mathbf{X} \mathbf{w}(k)) = \mathbf{w}(j)^T \mathbf{X}^T \mathbf{X} \mathbf{w}(k) = \mathbf{w}(j)^T \boldsymbol{\lambda}(k) \mathbf{w}(k) = \boldsymbol{\lambda}(k)^T \mathbf{w}(j) \mathbf{w}(k)$$

3.

Dimensionality reduction The transformation $P = X W$ maps a data vector $x(i)$ from an original space of x variables to a new space of p variables which are uncorrelated over the dataset. To non-dimensionalize the centered data, let X_c represent the characteristic values of data vectors X_i , given by:

4.

Software/source code ALGLIB – a C++ and C# library that implements PCA and truncated PCA
 Analytica – The built-in EigenDecomp function computes principal components. ELKI – includes PCA
 for projection, including robust variants of PCA, as well as PCA-based clustering algorithms. Gretl –
 principal component analysis can be performed either via the `pca` command or via the `princomp()`
 function. Julia – Supports PCA with the `pca` function in the MultivariateStats package. KNIME – A
 java based nodal arranging software for Analysis, in this the nodes called PCA, PCA compute, PCA

Apply, PCA inverse make it easily. Maple (software) – The PCA command is used to perform a principal component analysis on a set of data. Mathematica – Implements principal component analysis with the PrincipalComponents command using both covariance and correlation methods. MathPHP – PHP mathematics library with support for PCA. MATLAB – The SVD function is part of the basic system. In the Statistics Toolbox, the functions princomp and pca (R2012b) give the principal components, while the function pcares gives the residuals and reconstructed matrix for a low-rank PCA approximation. Matplotlib – a Python library that has a PCA package in the .mlab module. mlpack – Provides an implementation of principal component analysis in C++. mrmath – A high performance math library for Delphi and FreePascal can perform PCA; including robust variants. NAG Library – Principal components analysis is implemented via the g03aa routine (available in both the Fortran versions of the Library). NMath – Proprietary numerical library containing PCA for the .NET Framework. GNU Octave – Free software computational environment mostly compatible with MATLAB, the function princomp gives the principal component. OpenCV Oracle Database 12c – Implemented via DBMS_DATA_MINING.SVDS_SCORING_MODE by specifying setting value SVDS_SCORING_PCA. Orange (software) – Integrates PCA in its visual programming environment. PCA displays a scree plot (degree of explained variance) where user can interactively select the number of principal components. Origin – Contains PCA in its Pro version. Qlucore – Commercial software for analyzing multivariate data with instant response using PCA. R – Free statistical package, the functions princomp and prcomp can be used for principal component analysis; prcomp uses singular value decomposition which generally gives better numerical accuracy. Some packages that implement PCA in R, include, but are not limited to: ade4, vegan, ExPosition, dimRed, and FactoMineR. SAS – Proprietary software; for example, see. scikit-learn – Python library for machine learning which contains PCA, Probabilistic PCA, Kernel PCA, Sparse PCA and other techniques in the decomposition module. Scilab – Free and open-source, cross-platform numerical computational package, the function princomp computes principal component analysis, the function pca computes principal component analysis with standardized variables. SPSS – Proprietary software most commonly used by social scientists for PCA, factor analysis and associated cluster analysis. Weka – Java library for machine learning which contains modules for computing principal components.

5.

Principal component analysis creates variables that are linear combinations of the original variables. The new variables have the property that the variables are all orthogonal. The PCA transformation can be helpful as a pre-processing step before clustering. PCA is a variance-focused approach seeking to reproduce the total variable variance, in which components reflect both common and unique variance of the variable. PCA is generally preferred for purposes of data reduction (that is, translating variable space into optimal factor space) but not when the goal is to detect the latent construct or factors. Factor analysis is similar to principal component analysis, in that factor analysis also involves linear combinations of variables. Different from PCA, factor analysis is a correlation-focused approach seeking to reproduce the inter-correlations among variables, in which the factors "represent the common variance of variables, excluding unique variance". In terms of the correlation matrix, this corresponds with focusing on explaining the off-diagonal terms (that is, shared co-variance), while PCA focuses on explaining the terms that sit on the diagonal. However, as a side result, when trying to reproduce the on-diagonal terms, PCA also tends to fit relatively well the off-diagonal correlations. Results given by PCA and factor analysis are very similar in most situations, but this is not always the case, and there are some problems where the results are significantly different. Factor analysis is

generally used when the research purpose is detecting data structure (that is, latent constructs or factors) or causal modeling. If the factor model is incorrectly formulated or the assumptions are not met, then factor analysis will give erroneous results.